Errata

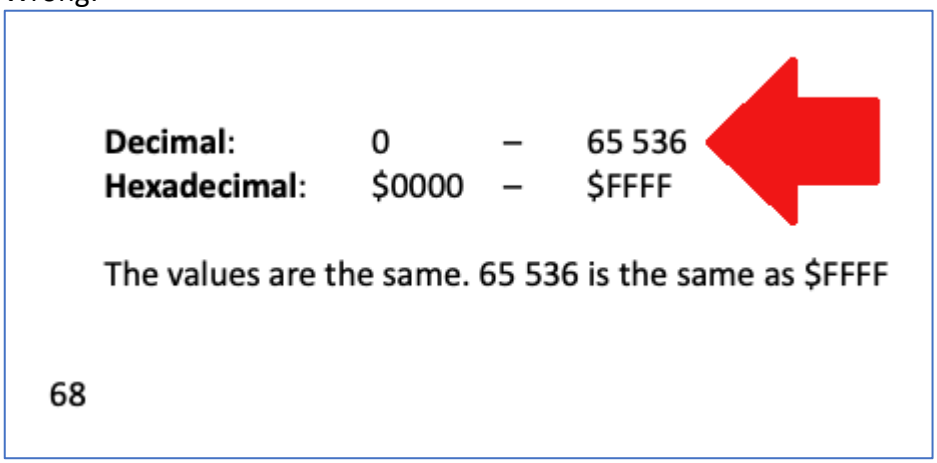
(Book Corrections)

The following mistakes were corrected and published as a new book on 5 November 2025.

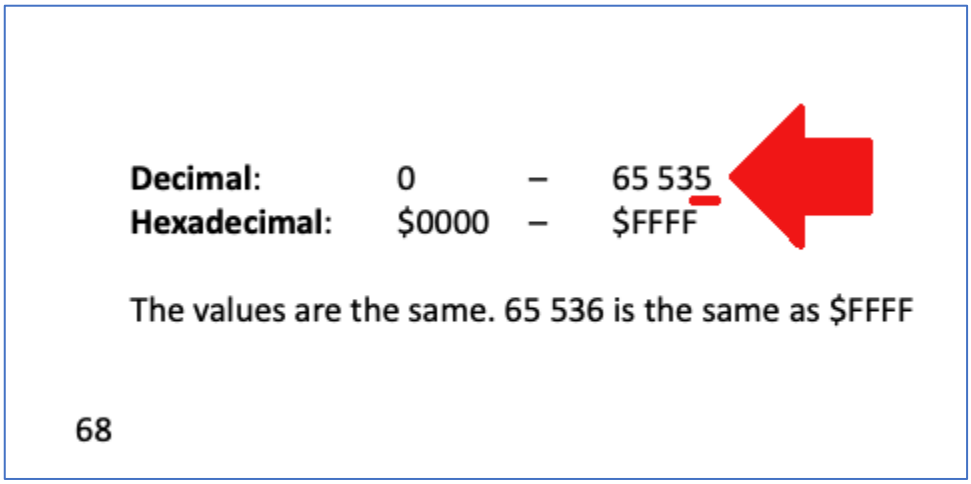
Page 68

- At the bottom of the page, the number 65 536 should be 65 53**5**.

Wrong:



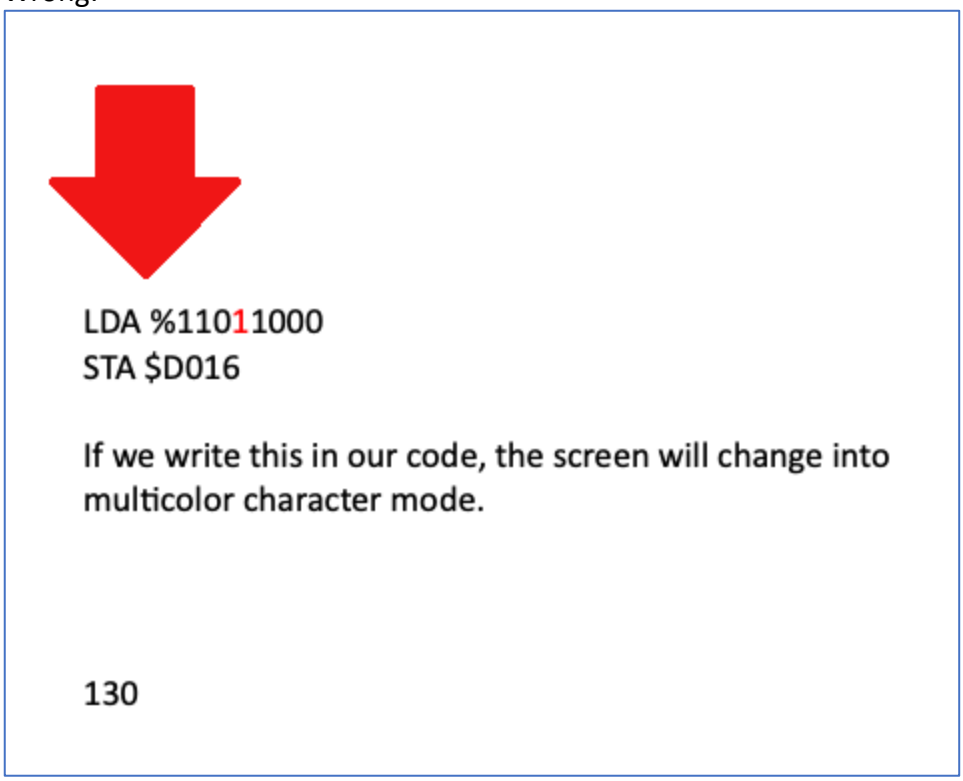
Correct:



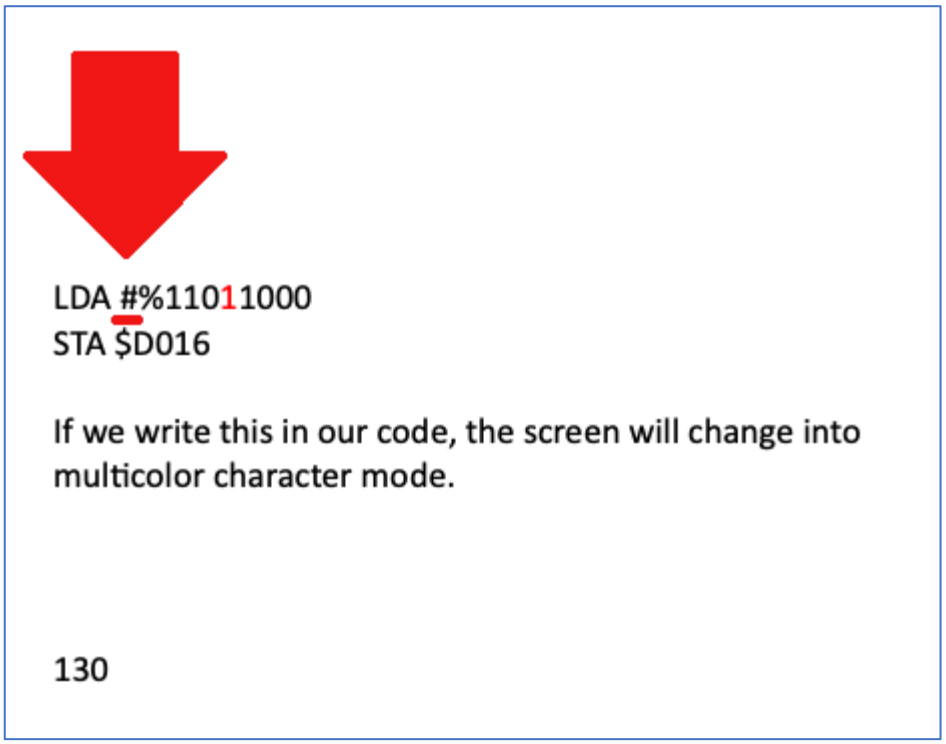
Page 130

- Towards the bottom of the page there is a hash symbol missing in front of the percent sign. It says LDA %11011000. It should be LDA **#**%11011000.

Wrong:



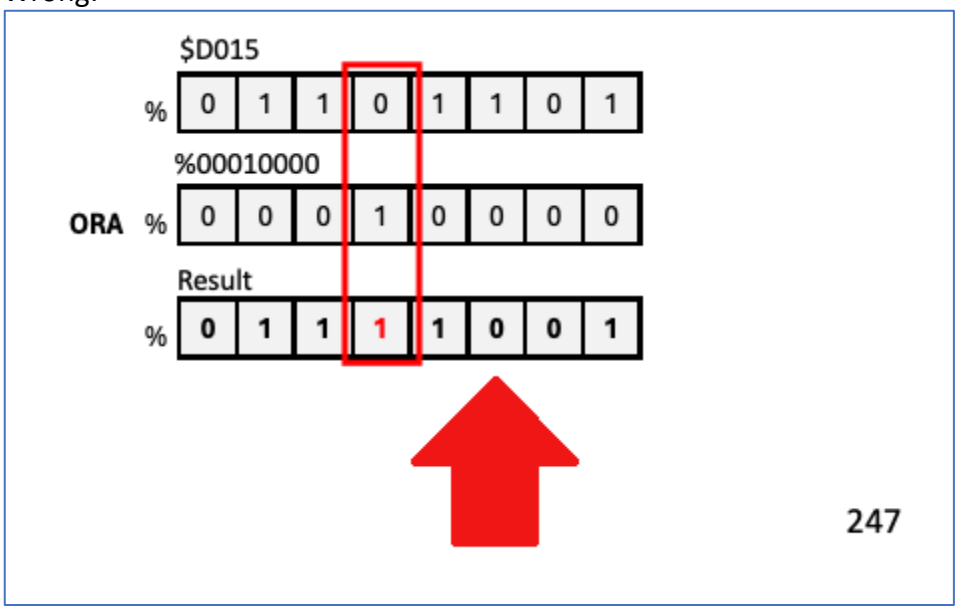
Correct:



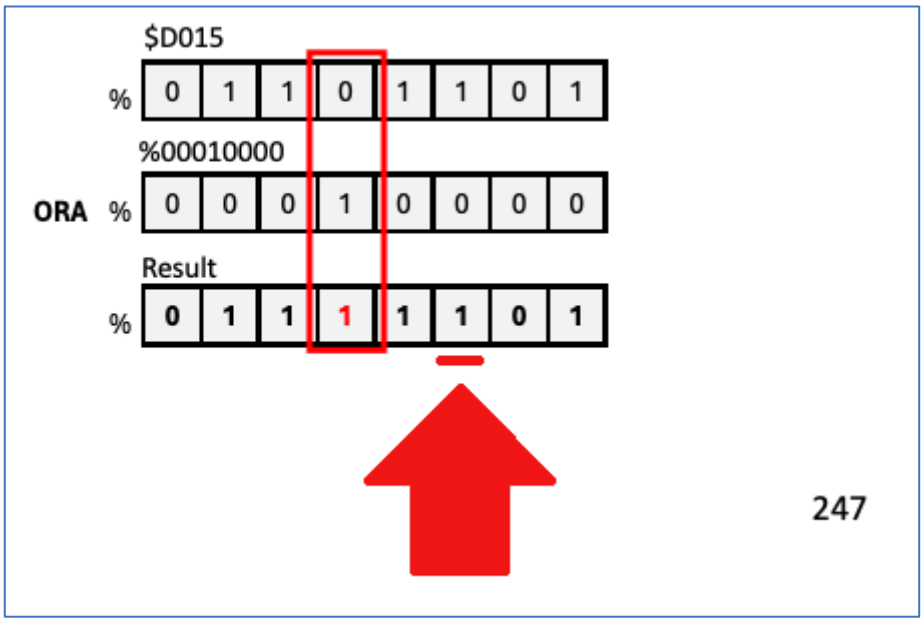
Page 247

- At the bottom of the page there is a figure showing how OR is used with binary numbers. At the bottom (Result) row there is a 0 where it should be a 1.

Wrong:



Correct:



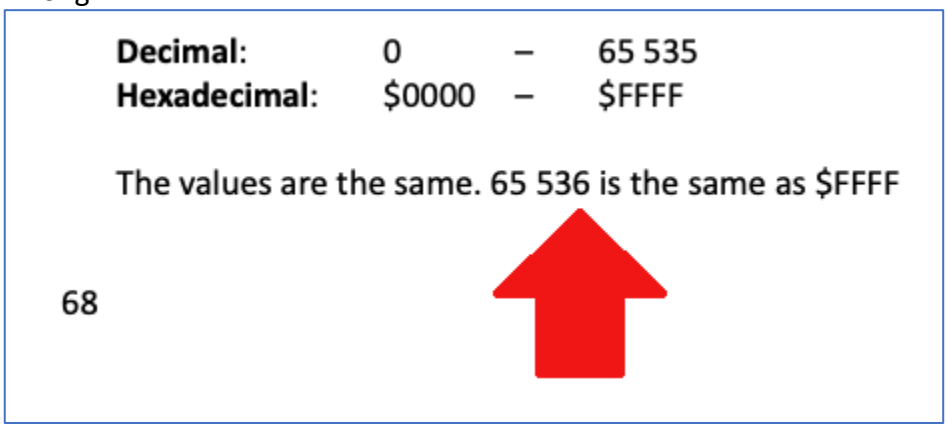
The following mistakes were corrected and published as a new book on 8 November 2025.

Page 68

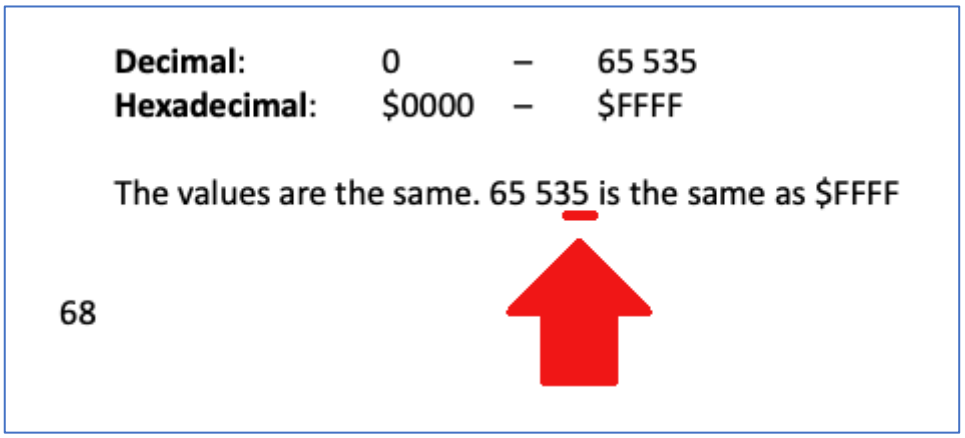
I can't believe I didn't see this during the previous correction, but the same mistake was present just two lines below the previous. This is the same mistake.

- At the bottom of the page, the number 65 536 should be 65 53**5**.

Wrong:



Correct:



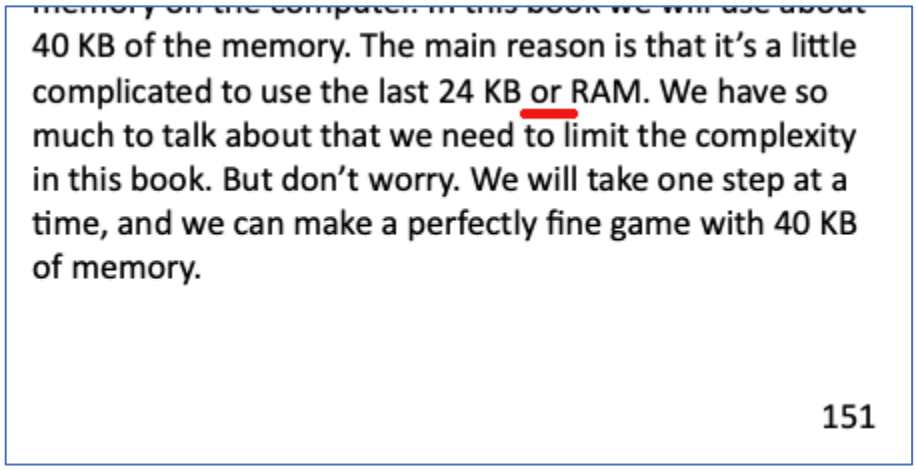
The following mistakes were corrected and published as a new book on 6 March 2026.

Page 151

There is a misspelling towards the bottom of the page.

It says: "… 24 KB or RAM."

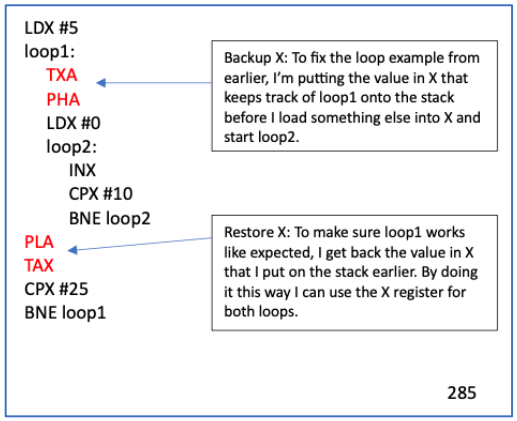
It should be: "… 24 KB of RAM."

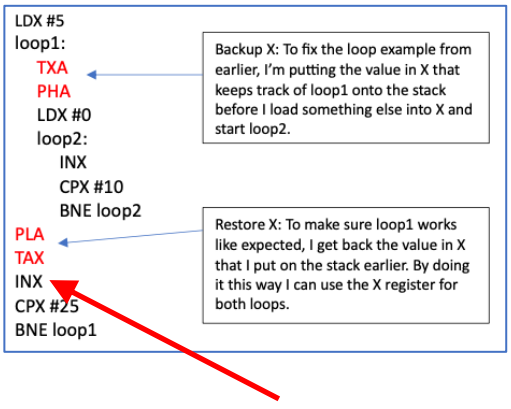

40 KB of the memory. The main reason is that it's a little complicated to use the last 24 KB or RAM. We have so much to talk about that we need to limit the complexity in this book. But don't worry. We will take one step at a time, and we can make a perfectly fine game with 40 KB of memory.

151

Page 285

The outer loop is missing an increment for the X register (INX).



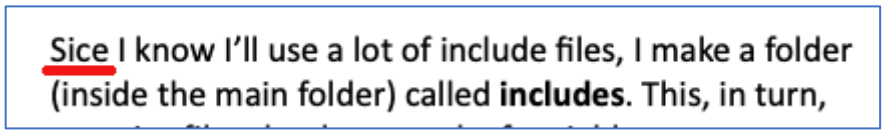


Page 315

Around the middle of the page there is a spelling mistake.

It says: “Sice I know…”

It should be: “Since I know…”

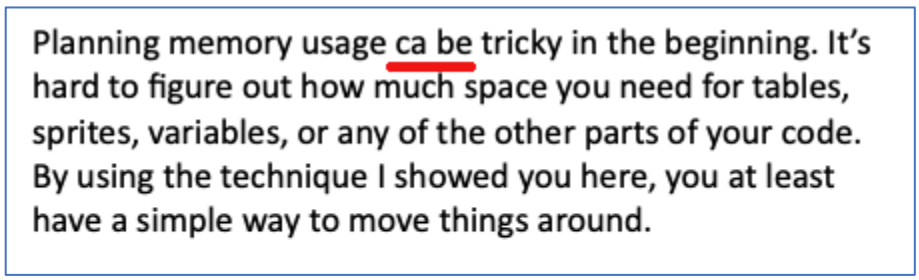

Sice I know I'll use a lot of include files, I make a folder (inside the main folder) called **includes**. This, in turn,

Page 317

Towards the bottom of the page there is a spelling mistake.

It says: “…ca be…”

It should be: “…can be…”

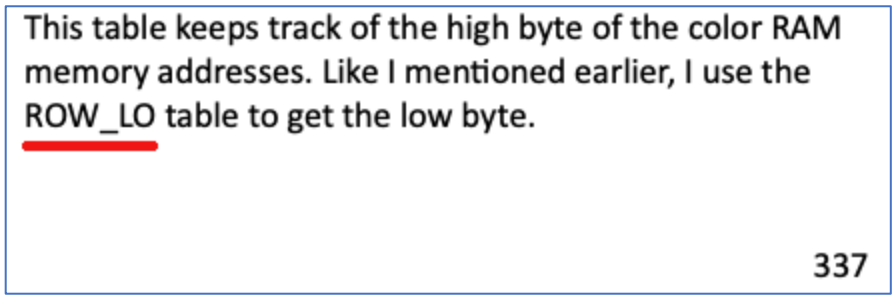

Planning memory usage ca be tricky in the beginning. It's hard to figure out how much space you need for tables, sprites, variables, or any of the other parts of your code. By using the technique I showed you here, you at least have a simple way to move things around.

Page 337

Towards the bottom of the page there is a minor mistake.

It says: “ROW_LO”

It should be: “Row_LO”


This table keeps track of the high byte of the color RAM memory addresses. Like I mentioned earlier, I use the ROW_LO table to get the low byte.

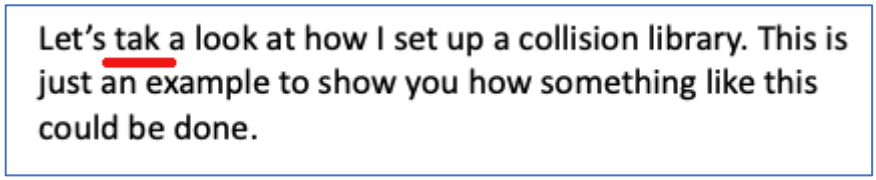
337

Page 369

On the first line of the page there is a spelling mistake.

It says: “Let’s tak a look…”

It should be: “Let’s take a look…”

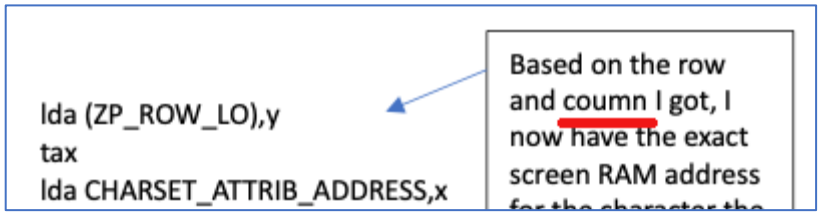


Page 370

In the upper box on the right there is a spelling mistake.

It says: “…coumn…”

It should be: “…column…”

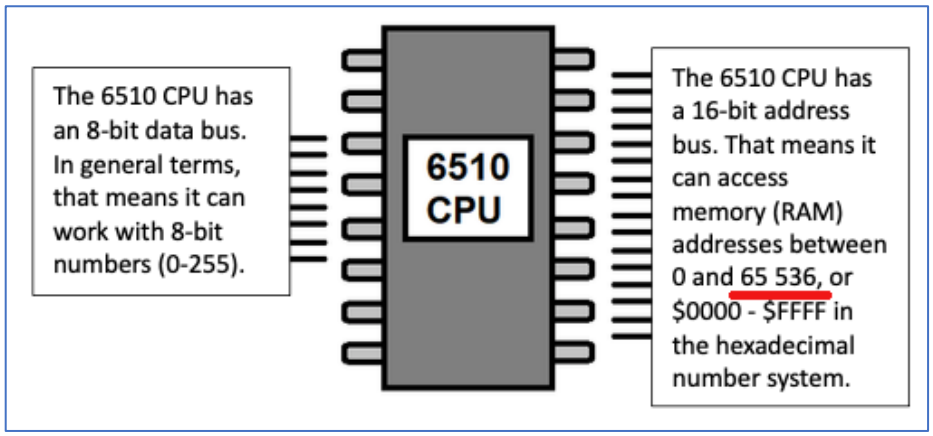


Page 9

In the box on the right there is a mistake. The same mistake that has been haunting me since I started doing the book corrections.

It says: “65 536”

It should be: “65 535”



Page 23

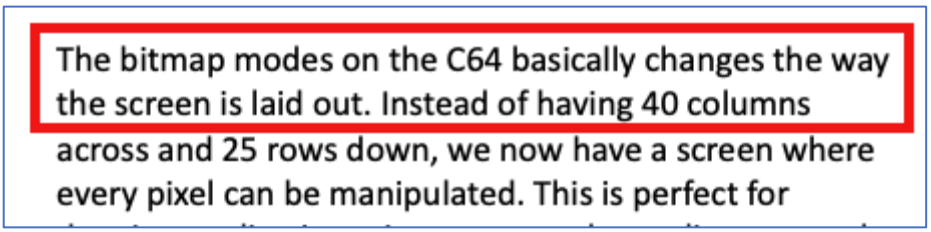
A minor rewrite of the first sentence.

Previous sentence:

“The bitmap modes on the C64 basically changes the way…”

Current sentence:

“The various bitmap modes on the C64 change the way…”


The bitmap modes on the C64 basically changes the way the screen is laid out. Instead of having 40 columns across and 25 rows down, we now have a screen where every pixel can be manipulated. This is perfect for

Page 33

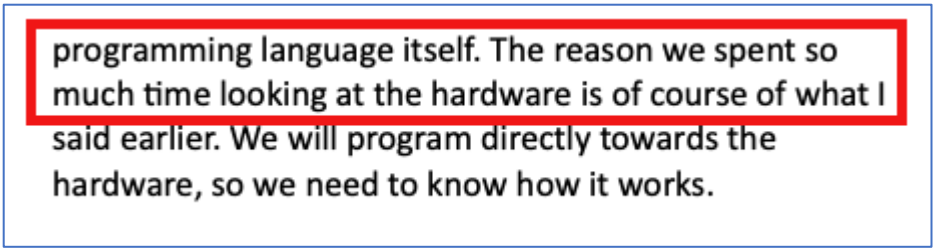
A minor rewrite of a sentence in the first paragraph.

Previous sentence:

“The reason we spent so much time looking at the hardware is **of course** of what…”

Current sentence:

“The reason we spent so much time looking at the hardware is **because** of what…”


programming language itself. The reason we spent so much time looking at the hardware is of course of what I said earlier. We will program directly towards the hardware, so we need to know how it works.

Page 39

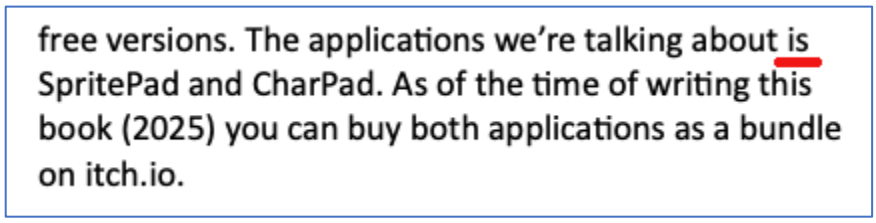
A minor rewrite of a sentence in the middle of the page.

Previous sentence:

“The applications we’re talking about **is** SpritePad and CharPad.”

Current sentence:

“The applications we’re talking about **are** SpritePad and CharPad.”

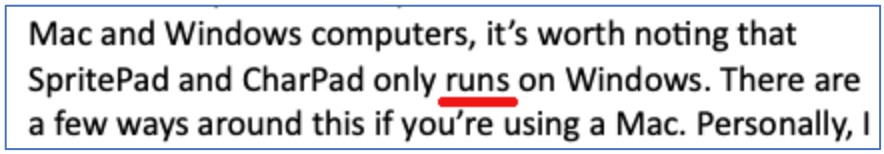

free versions. The applications we’re talking about is SpritePad and CharPad. As of the time of writing this book (2025) you can buy both applications as a bundle on itch.io.

Page 40

On line 6 there is a spelling mistake.

It says: “…SpritePad and CharPad only **runs** on Windows.”

It should be: “…SpritePad and CharPad only **run** on Windows.”

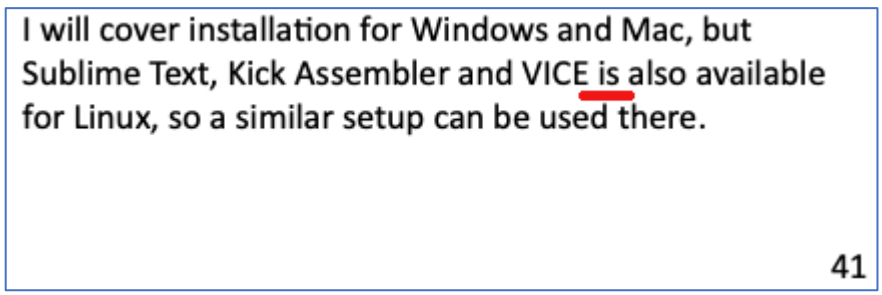

Mac and Windows computers, it’s worth noting that SpritePad and CharPad only runs on Windows. There are a few ways around this if you’re using a Mac. Personally, I

Page 41

Towards the bottom of the page there is an error.

It says: “…Sublime Text, Kick Assembler and VICE **is** also...”

It should be: “…Sublime Text, Kick Assembler and VICE **are** also...”


I will cover installation for Windows and Mac, but Sublime Text, Kick Assembler and VICE is also available for Linux, so a similar setup can be used there.

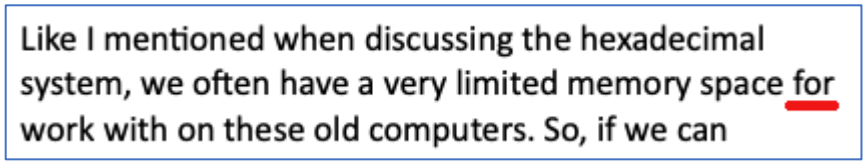
41

Page 73

There is an error at the beginning of the second paragraph.

It says: "…a very limited memory space **for** work with..."

It should be: "…a very limited memory space **to** work with..."

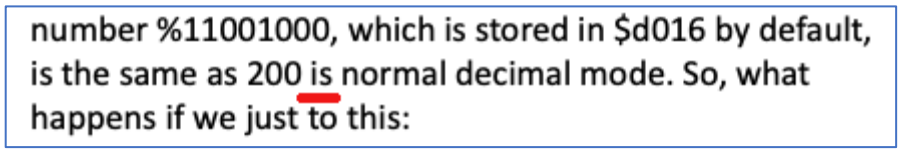


Page 130

Towards the end of the first paragraph there is a spelling mistake.

It says: "…the same as 200 **is** normal decimal mode…"

It should be: "…the same as 200 **in** normal decimal mode…"

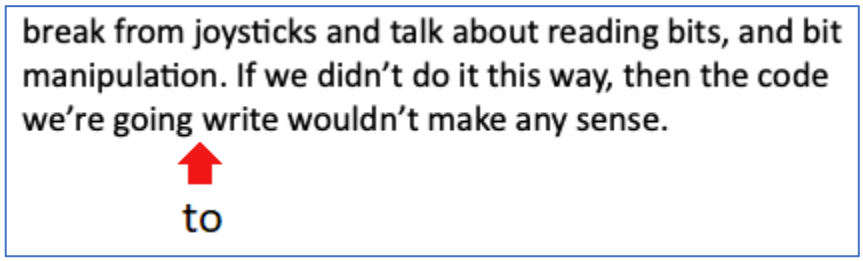


Page 242

There is an error in the last sentence of the paragraph on the top.

It says: "…we're going write wouldn't make any…"

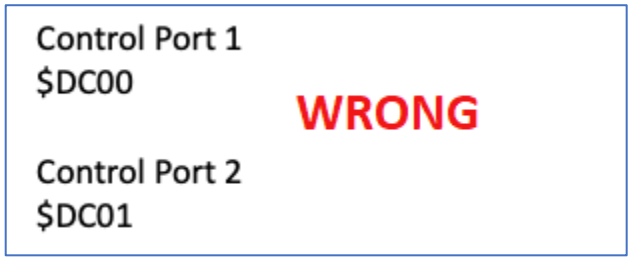
It should be: "…we're going **to** write wouldn't make any…"



Page 249

There is a mistake with the names of the memory addresses for the joysticks. They are switched around. Control Port 1 and Control Port 2 are mixed up.

It says:



It should be:

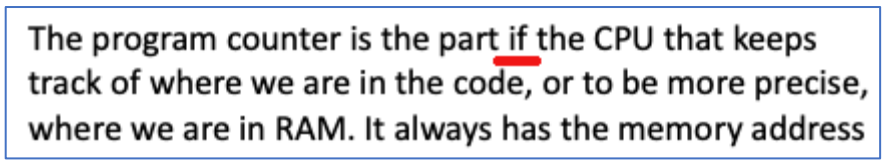


Page 435

In the first sentence of the second paragraph there is a spelling mistake.

It says: “The program counter is the part **if** the CPU...”

It should be: “The program counter is the part **of** the CPU...”



Page 2

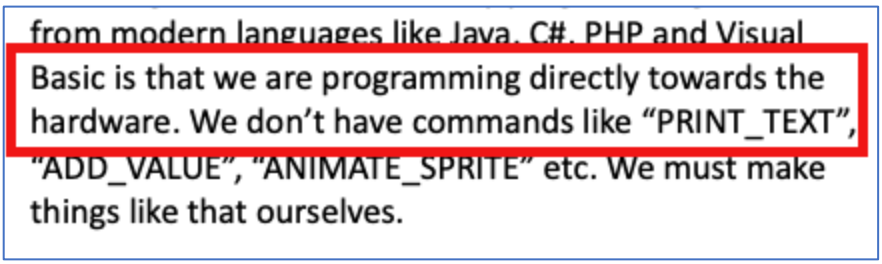
A rewrite of a sentence towards the end of the second paragraph.

Previous sentence:

"…is that we are programming directly towards the hardware..."

Current sentence:

"…is that we are programming the hardware directly..."


from modern languages like Java, C#, PHP and Visual Basic is that we are programming directly towards the hardware. We don't have commands like "PRINT_TEXT", "ADD_VALUE", "ANIMATE_SPRITE" etc. We must make things like that ourselves.

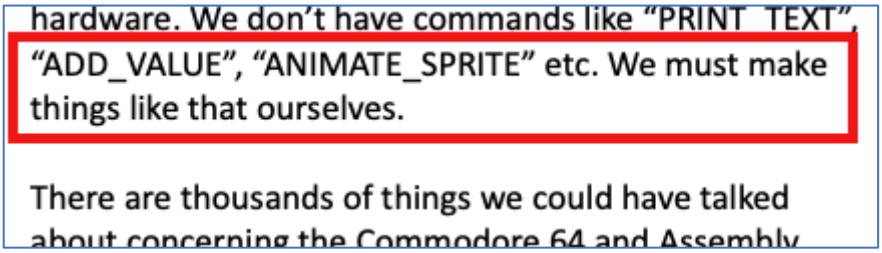
A rewrite of the last sentence in the same paragraph.

Previous sentence:

"We must make things like that ourselves."

Current sentence:

"We must write our own routines for those purposes."


hardware. We don't have commands like "PRINT_TEXT", "ADD_VALUE", "ANIMATE_SPRITE" etc. We must make things like that ourselves.

There are thousands of things we could have talked about concerning the Commodore 64 and Assembly

Page 3

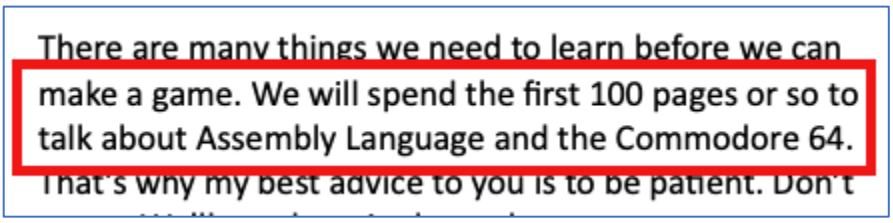
A rewrite of a sentence in the middle of the last paragraph, towards the bottom of the page.

Previous sentence:

“We will spend the first 100 pages or so to talk about…”

Current sentence:

“We will spend the first 100 pages or so talking about…”

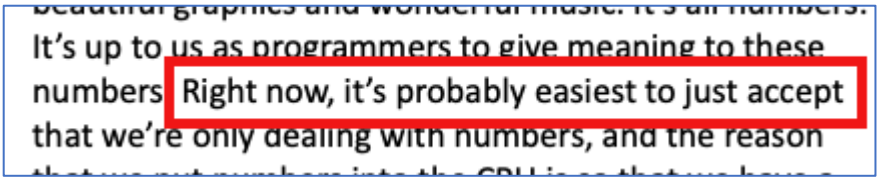


Page 7

There is an error in the middle of the page.

It says: “Right now, it’s probably easiest **to just**…”

It should be: “Right now, it’s probably easiest **just to**…”



Page 8

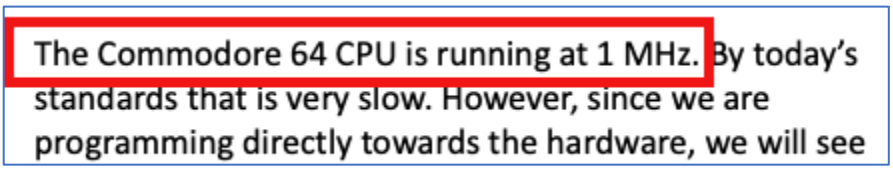
A rewrite of the first sentence in the second paragraph.

Previous sentence:

“The Commodore 64 CPU is running at 1 MHz.”

Current sentence:

“The Commodore 64 CPU runs at 1 MHz.”



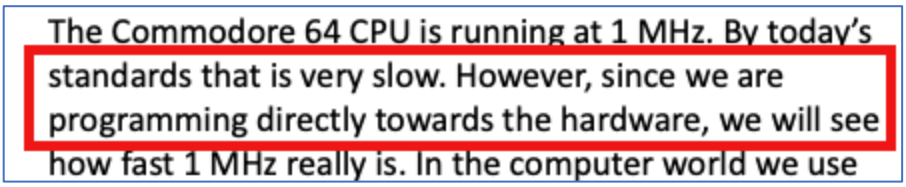
A rewrite of a sentence on the second line in the same paragraph.

Previous sentence:

“However, since we are programming directly towards the hardware…”

Current sentence:

“However, since we are programming the hardware directly…”



A rewrite of the last sentence in the same paragraph.

Previous sentence:

“Our Assembly code will run hundreds of times faster than if the code was written in the BASIC programming language.”

Current sentence:

“Our Assembly code will run hundreds of times faster than if the code had been written in BASIC.”

Page 11

A rewrite of the first sentence in the first paragraph.

Previous sentence:

“We put numbers inside the CPU and maybe do some simple math on it, and then store the result in memory.”

Current sentence:

“We put numbers inside the CPU registers and do some simple math on it. Then, we can store the result in memory.”

Page 13

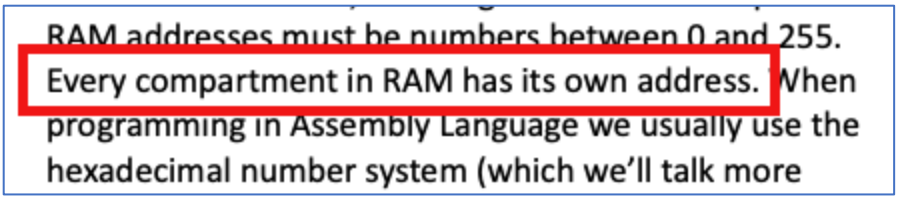
A rewrite of a sentence in the middle of the page.

Previous sentence:

“Every compartment in RAM has its own address.”

Current sentence:

“Every compartment in RAM has its own **unique address**.”


RAM addresses must be numbers between 0 and 255.
Every compartment in RAM has its own address. When
programming in Assembly Language we usually use the
hexadecimal number system (which we'll talk more

Page 21

A rewrite of a sentence in the middle of the second paragraph.

Previous sentence:

“The tradeoff is that the width of the pixels that makes up the character is doubled.”

Current sentence:

“The tradeoff is that the width of the pixels that make up each character become doubled.”

Page 32

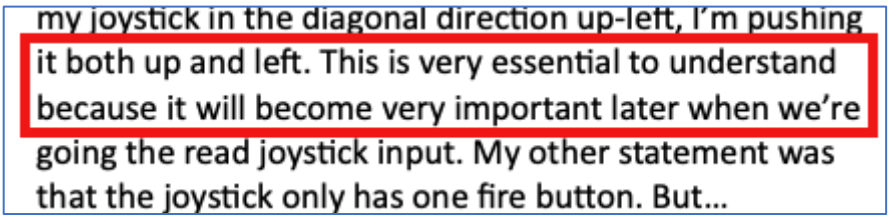
A rewrite of a sentence in the middle of the paragraph.

Previous sentence:

“This is very essential to understand because…”

Current sentence:

“It is essential to understand this as…”


my joystick in the diagonal direction up-left, I'm pushing it both up and left. This is very essential to understand because it will become very important later when we're going the read joystick input. My other statement was that the joystick only has one fire button. But…

Page 33

A rewrite of the last sentence in the first paragraph.

Previous sentence:

“We will program directly towards the hardware…”

Current sentence:

“We will program the hardware directly…”

Page 34

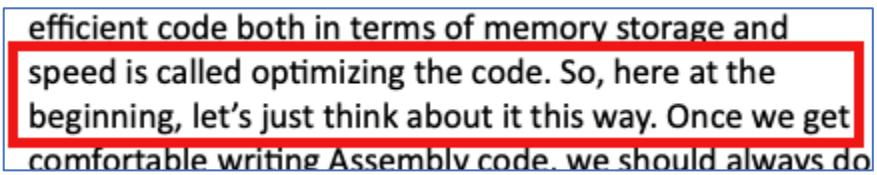
A rewrite of a sentences towards the end of the paragraph.

Previous sentence:

“So, here at the beginning, let's just think about it this way.”

Current sentence:

“Therefore, to start with, let's just think about it this way.”


efficient code both in terms of memory storage and speed is called optimizing the code. So, here at the beginning, let's just think about it this way. Once we get comfortable writing Assembly code, we should always do

A rewrite of a sentence almost at the end of the paragraph on page 34.

Previous sentence:

“But like I said. It takes some time…”

Current sentence:

“But as I have said, it takes some time…”

Page 67

Several rewrites of sentences.

Previous:
“When will we get to actual programming..?”
Current:
“When will be start programming the Commodore 64...?”

Previous:
“…journey **for** every page.”
Current:
“…journey **with** every page.”

Previous:
“…program directly towards the hardware.”
Current:
“…program the hardware directly.”

Previous:
“One thing that often **confuse** beginners…”
Current:
“One thing that often **confuses** beginners…”

Previous:
“…unless I have a reason…”
“…unless I have a **specific** reason…”

Page 69

Two rewrites of sentences.

Previous:
"…create **a** messy looking code…"
Current:
"…create messy looking code…"

Previous:
"…may perhaps see a value in using it."
Current:
"…might see the reason for using it."

Page 72

A punctuation mistake in the box on the right.

"…(in other words 0)." – Should be "…(in other words**,** 0)."

Page 75

A mistake in the last sentence of the second paragraph.

"…make sure to NOT use a…" – Should be "…make sure NOT to use a…"

Page 76

A punctuation mistake in the first sentence of the first paragraph.

"…variable is**. B**ut let's not…" – Should be "…variable is**, b**ut let's not…"

Page 81

A minor mistake in the first sentence of the first paragraph.

"…Commodore 64s memory…" – Should be "…Commodore 64's memory…"

Page 87

A rewrite of the last sentence on the bottom of the page.

Previous:
"Often times it's not noticeable…"
Current:
"Usually it's not noticeable…"

Page 91 & 92 and all other places

'Basically' – Should say 'essentially' as basically is considered bad English.

Page 108

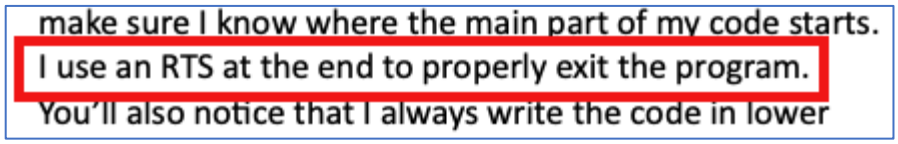
A rewrite of a sentence in the middle of the page.

Previous sentence:

"I use an RTS at the end to properly exit the program."

Current sentence:

"I use an RTS at the end of the program code to exit the program properly."



Page 109

A minor rewrite of a sentence in the middle of the second paragraph.

Previous sentence:
"So, even though we just wrote our first lines of code…"

Current sentence:
"So, even though we just wrote our first **few** lines of code…"

Page 110

A rewrite of the last sentence of the first paragraph.

Previous sentence:
"…it will gradually be clearer for each exercise we do."

Current sentence:
"…it will gradually become clearer as we progress through the exercises."

Page 111

A rewrite of a sentence towards the beginning of the first paragraph.

Previous sentence:
“Like we talked about at the beginning of the book…”

Current sentence:
“As explained at the beginning of the book…”

In the middle of the second paragraph, there is a space missing between the word **built** and the hyphen:



“built- in” – Should be “built-in”

Page 112

Punctuation mistake at the beginning of the first paragraph.

”…lower right corner.”
“…lower right corner?”

A rewrite of the last sentence on the page.

Previous sentence:
“So, in the end, that’s what…”

Current sentence:
“Finally, that’s what…”

Page 113

A minor change in a sentence towards the bottom of the page.

Previous sentence:
“…I “tweak” the memory…”

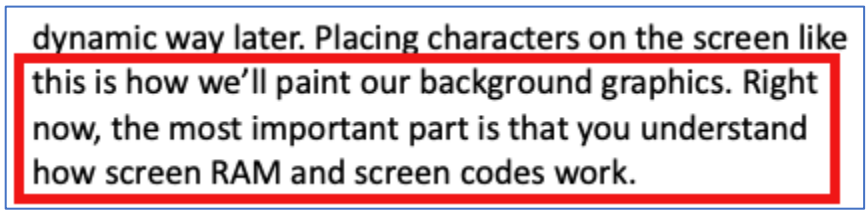
Current sentence:
“…I “adapt” the memory…”

Page 115

A rewrite of a sentence towards the beginning of the first paragraph.

Previous sentence:
“Right now, the most important part is that you understand how screen RAM and screen codes work.”

Current sentence:
“Consequently, it is essential you now understand how both screen RAM and screen codes work.”


dynamic way later. Placing characters on the screen like this is how we'll paint our background graphics. Right now, the most important part is that you understand how screen RAM and screen codes work.

A rewrite of the first sentence of the last paragraph on the page.

Previous sentence:
“We don’t have to do this, but often times it makes the code much easier to read.”

Current sentence:
“We don’t have to do this, but it makes the code far easier to read.”

Page 116

A minor correction towards the bottom of the page. A missing ‘s’ after graphic.

“…we can make our own graphic characters.”
“…we can make our own graphic**s** characters.”

Page 117

A minor mistake in the last sentence on the page.

“This is an example where I change…”
“This is an example **of** where I change…”

Page 119

A minor rewrite of the last sentence in the first paragraph.

"…to understand how RAM and ROM works on the C64."
"…to understand how **both** RAM and ROM **work** on the C64."

A minor punctuation mistake towards the end of the last paragraph on the page.

"That doesn't make any sense. Does it?"
"That doesn't make any sense, does it?"

Page 120

A minor punctuation mistake towards the end of the page.

"…it somehow needs to be in the Commodore 64 memory."
"…it somehow needs to be in the Commodore 64's memory."

Page 121

A minor punctuation mistake and a missing 's' in the first sentence of the page.

"…is a part of the Commodore 64 RAM."
"…is a part of the Commodore 64's RAM."

A rewrite of the last sentence on the page.

Previous sentence:
"That is something we need to look closer at in this chapter."

Current sentence:
"That is something we will be examining in greater detail in this chapter."

Page 123

Minor rewrites of the first and second sentences of the page.

Previous:
"Just like with the characters…"
Current:
"Just as with the characters…"

Previous:
"And just like with the screen codes…"
Current:
"And just as with the screen codes…"

Page 125

A minor correction of the first sentence below the image.

From:
“So, now we know that each character on the screen can have **their** own color.”

To:
“So, now we know that each character on the screen can have its own color.”

A small rewrite two lines further down:

From:
“…high resolution mode **like** we’re doing here.”

To:
“…high resolution mode as we’re doing here.”

Page 127

A rewrite of the last sentence of the page.

Previous sentence:
“The important part is that you understand how screen RAM and color RAM works.”

Current sentence:
“The important concept here is that you understand how screen RAM and color RAM both work.”

Page 128

A rewrite of the last sentence of the page.

Previous sentence:
“…not to go too much **into** detail about this.”

Current sentence:
“…not to go **into** too much detail about this.”

Page 129

A minor change of the sentence right before the table on the middle of the page.

From:
“They are numbered from 7 on the left and goes down to 0 on the right.”

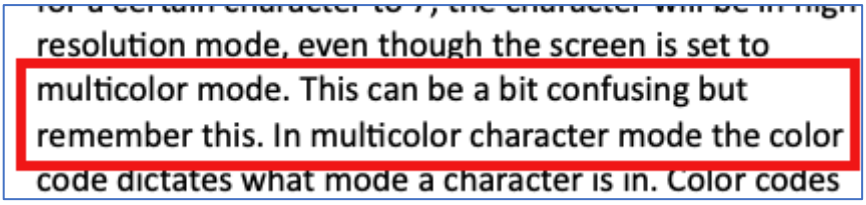
To:
“They are numbered from 7 on the left, decrementing with each bit to 0 on the right.”

Page 132

Punctuation mistakes in the bottom half of the first paragraph.

“This can be a bit confusing but remember this.”

“This can be a bit confusing, but remember this:”



Page 136

Correction in the paragraph at the bottom of the page.

From:
“…and multicolor colors to better **understand**…”

To:
“…and multicolor colors to **understand** better…”

Page 142

Correction of the first sentence on the page.

From:
“So, now the obvious question, why would…”

To:
“So, now the obvious question: Why would…”

Page 146

A minor change of the first sentence of the page.

From:
“We have taken a very…”

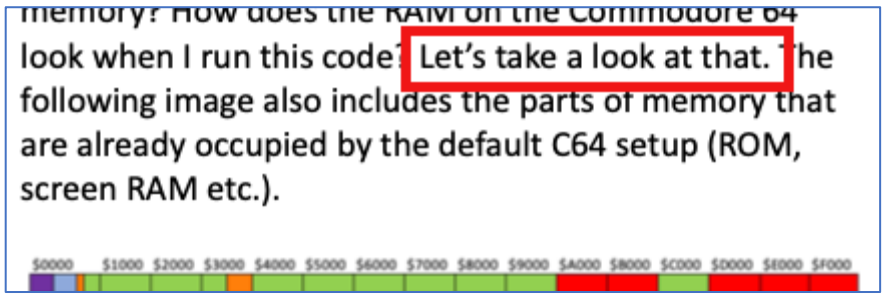
To:
“We have just taken a very…”

Page 150

A rewrite of a sentence around the middle of the page.

Previous sentence:
“Let’s take a look at that.”

Current sentence:
“Let’s examine what the memory layout looks like.”



Page 151

Punctuation mistake in the last sentence of the page.

From:
“But don’t worry. We will…”

To:
“But don’t worry, we will…”

Page 158

A rewrite of a sentence in the middle of the first paragraph.

From:
“…pixel grid makes it hard too fully understand how this is structured…”
To:
“…pixel grid makes it too hard to understand fully how this is structured…”

Page 164

A minor spelling mistake in the first sentence of the last paragraph on the page.

From:
“The unique sprite colors for each sprite does not change…”

To:
“The unique sprite color for each sprite does not change…”

A rewrite of the last sentence of the page.

From:
“…(like we’ve seen before)…”

To:
“…(as we’ve seen earlier)…”

Page 165

A minor punctuation mistake, as well as a minor rewrite of the first sentence of the first paragraph.

From:
“…multicolor mode we need to both **set** the color…”

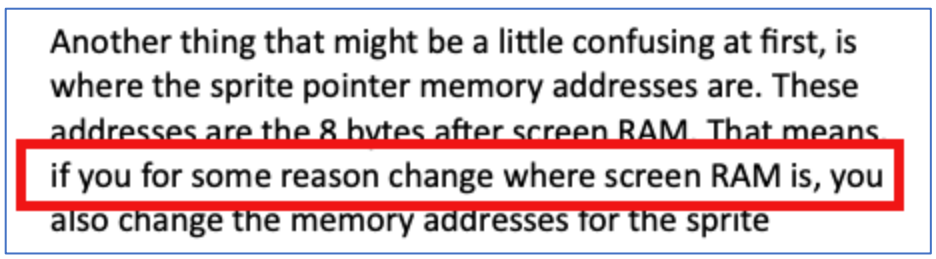
To:
“…multicolor mode, we need to **set** both the color…”

Page 168

A rewrite of a sentence in the first paragraph.

From:
“…If you for some reason change where…”

To:
“…If, for some reason, you change where…”


Another thing that might be a little confusing at first, is where the sprite pointer memory addresses are. These addresses are the 8 bytes after screen RAM. That means if you for some reason change where screen RAM is, you also change the memory addresses for the sprite

Page 173

Minor correction of the first sentence of the second paragraph.

From:
“As you may have guessed, each sprite has their own memory…”

To:
“As you may have guessed, each sprite has its own memory…”

In the middle of the second paragraph:

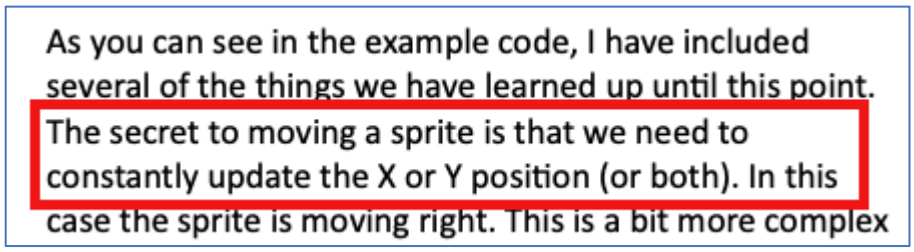
“…for the Y coordinate…” – Should be “…for its Y coordinate…”

Page 179

Rewrite of a sentence towards the beginning of the paragraph.

From:
“…is that we need to constantly update the X or Y position (or both).”

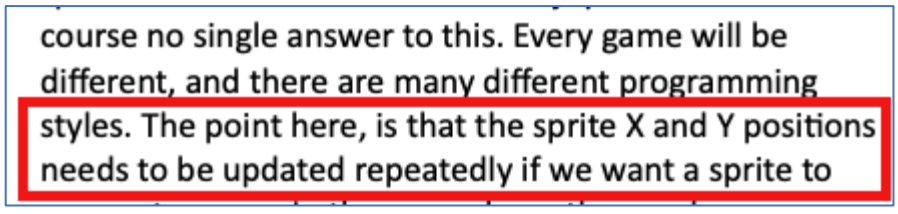
To:
“…is that we need to update the X or Y position (or both) constantly.”


As you can see in the example code, I have included several of the things we have learned up until this point. The secret to moving a sprite is that we need to constantly update the X or Y position (or both). In this case the sprite is moving right. This is a bit more complex

Punctuation mistake slightly above the middle of the page.

From:
“…the sprite X and Y positions needs to be…”

To:
“…the sprite’s X and Y positions need to be…”


course no single answer to this. Every game will be different, and there are many different programming styles. The point here, is that the sprite X and Y positions needs to be updated repeatedly if we want a sprite to

Page 183

Slight rewrite of the last sentence of the second paragraph.

From:
“…until it smoothly moves…”
To:
“…until it moves smoothly…”

Page 231

Slight correction of the first sentence of the second paragraph.

From:
“Like we saw in SpritePad…”
To:
“As we saw in SpritePad…”

Page 242

Rewrite of the first sentence on the page.

From:
“Now we have gotten…”
To:
“Now we have reached…”

Page 243

Slight rewrite of a sentence towards the top of the page.

From:
“But before we can properly **read** joystick input…”
To:
“But before we can **read** joystick input properly…”

Page 258

Rewrite of last sentence of the page.

From:
“Be patient and stick with it, and I guarantee that you’ll eventually get the hang of it.”
To:
“If you are patient and stick with it, I guarantee you’ll eventually get the hang of it.’

Page 260

Spelling mistake in the last sentence of the first paragraph.

From:
“…same, the numers just look…”

To:
“…same, the numbers just look…”

Page 284

Punctuation mistake slightly above the middle of the second paragraph.

From:
“…1 to 2, 2 to 3, and…”

To:
“…1 to 2, 2 to 3 and…”

Page 289

Slight rewrite of last sentence in fourth paragraph.

From:
“…how you can make a routine to easily **write** text…”
To:
“…how you can make a routine to **write** text easily…”

Page 317

Spelling mistake in first sentence of last paragraph.

From:
“Planning memory usage ca be tricky…”

To:
“Planning memory usage can be tricky…”